COM5231 · Introduction to Wireless and Ad Hoc Networking

Part 2 — Study Notes

Multiple Access · Spread Spectrum · OFDM/OFDMA · Error Detection and Correction

SUPPLEMENTARY content not found in the uploaded lecture notes

NOTES ERROR a mistake in the lecture notes, with the verified correction

EXAM how the topic is or could be examined

★ high-priority, frequently examined content

Exam preparation pack, built from the uploaded lecture materials and past papers

Part 2 — Detailed Study Notes: Topics 5–8

Topics in this part 5. Multiple Access Techniques (FDMA, TDMA, CDMA, SDMA) and Cellular Generations 6. Spread Spectrum (FHSS, DSSS, CDMA anti-jamming) 7. OFDM, OFDMA and SC-FDMA 8. Error Detection and Correction (parity, Internet checksum, Hamming code, CRC)

Source codes used in this part

Code	File
[MA p]	*Multiple access techniquesV2.pdf*, page p (scanned slides)
[AT]	*Fundamentals of Communications Access Technologies.pdf* (Frenzel article)
[MX]	*Multiplexing notes.pdf*
[SS]	*Spread spectrum.pdf* (Forouzan §6.2)
[OF]	*OFDM.pdf* (Beard & Stallings Ch. 8)
[ED]	*Error detection and correction_v2.docx* (worksheet)
[WC-S n] / [WC-B §]	*The Wireless Channel.pptx* / *Wireless Communication.pdf*
[BT]	Bluetooth PDF and slides

TOPIC 5 — MULTIPLE ACCESS TECHNIQUES

5.1 Introduction

"Sharing the limited bandwidth efficiently among many users is one of the main objectives of multiple access schemes" [MA p1].

Cellular systems divide a region into **cells**, and a mobile in each cell communicates with a **base station**. The main objective of cellular design is to **handle as many calls as possible (capacity) in a given bandwidth with reliability**. Success depends on the availability of **radio frequency spectrum**, a **finite natural resource** defined in terms of **bandwidth**. The technologies that let many users share a finite amount of spectrum simultaneously and efficiently are called **multiple access** [MA p2].

The choice of multiple access technology **can significantly enhance or lower the service quality** delivered to end users [MA p2].

The four multiple access techniques [MA p2]: **FDMA, TDMA, CDMA, SDMA**. The article [AT] adds **OFDMA** as a fifth, and mentions CSMA as "other methods".

Multiplexing vs multiple access: multiplexing creates channels in frequency, time or code (FDM, TDM, CDM) [MA p3]. **Access methods are multiplexing techniques that provide communication services to multiple users** in a single-bandwidth medium [AT].

Highway analogy [MA p3]: the medium is the **highway**, the users are **cars**, and interference is **accidents**. Separate lanes = SDMA or FDMA; cars taking turns on one lane = TDMA.

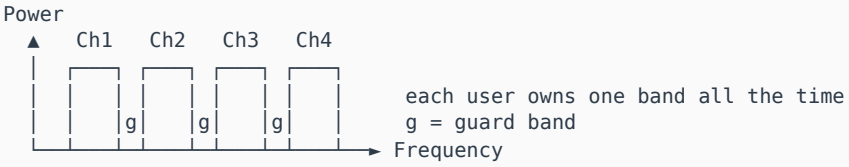
5.2 Key definitions

Term	Definition	Source
Multiple access	Allowing many mobile users to share a finite amount of radio spectrum simultaneously and efficiently, to maximise call-handling capacity in a given bandwidth	[MA p2]
FDMA	Radio spectrum divided into several frequency bands separated by **guard bands**; each band is used by one user simultaneously with the others	[MA p3]
Guard band	Unused frequency slots that separate neighbouring channels to minimise adjacent channel interference (crosstalk)	[MA p4]
TDMA	Several users share the same frequency band by dividing time into **slots** periodically allocated to each user	[MA p5]
CDMA	Transmissions **occupy the entire bandwidth at the same time** without interference, separated by **unique codes**	[MA p7]
Spread-spectrum signal	A signal with an **extra modulation** that expands its bandwidth **beyond what the data modulation requires**	[MA p7]
Chip rate	The rate of a spreading code (instead of "bit rate")	[MA p8]
Pseudorandom code sequence	Unique code shared by the mobile and the base station, used to spread and despread a user's data	[MA p8]
SDMA	Uses the **spatial dimension**: the same bandwidth is shared simultaneously by users in **different geographical locations** or **different beam directions**	[MA p9]

5.3 Frequency Division Multiple Access (FDMA) ★

Principle [MA p3–4, MX, AT]

- The **simplest** scheme; used for multiple access in **analogue** transmission.
- The spectrum is **divided into several frequency bands separated by guard bands**. Each band can be used simultaneously.
- The bandwidth is divided into channels and distributed among users, each with a **finite portion of bandwidth for permanent use**.
- **Only one user per channel**, who can use the channel **100% of the time**. Only the **frequency dimension** defines channels.
- **Guard bands** between adjacent channels minimise **crosstalk**.
- Each FDMA channel has **two frequencies**: a **forward channel** (base → mobile) and a **reverse channel** (mobile → base).
- While a user is "in conversation", **no other user can access the same spectrum**.
- Additional points [MX]: compatible with **both digital and analog** signals; needs **highly efficient filters**; **no timing issues** (unlike TDMA); **not prone to the near-far problem** of CDMA (thanks to frequency filtering); its weakness is **crosstalk** between frequencies.



Number of channels ★ [MA p4]

$$N = (Bt - 2Bg) / Bc$$

- **Bt** = allocated frequency spectrum (total bandwidth)
- **Bg** = guard band allocated **at the edge** of the spectrum
- **Bc** = channel bandwidth

Worked example FD1 SUPPLEMENTARY (standard textbook numbers, using the lecturer's formula)

A US AMPS-type system is allocated Bt = 12.5 MHz per direction, with Bg = 10 kHz at each edge and Bc = 30 kHz. How many channels?

1. Formula: N = (Bt − 2Bg)/Bc
2. Substitute in kHz: N = (12 500 − 2 × 10)/30 = 12 480/30
3. **N = 416 channels**

Worked example FD2 SUPPLEMENTARY

Bt = 5 MHz, Bg = 25 kHz, Bc = 200 kHz. N = (5000 − 50)/200 = 24.75 → **24 channels** (always round **down**: a partial channel can't be used).

Advantages of FDMA [MA p4]

1. **Continuous transmission**, so lower complexity than TDMA; synchronisation requirements are not severe.
2. **Simple hardware**: users are isolated with simple **band-pass filters**.
3. Fairly efficient with a **small user population** and **constant traffic**.
4. **No channel equalization** required.
5. Capacity can be increased by **reducing the information bit rate** and using efficient digital speech coding.

Disadvantages of FDMA [MA p4]

1. An idle channel **sits idle** and cannot be used by others → **inefficient use of spectrum**.
2. Needs **tight RF filtering** to minimise adjacent channel interference.
3. **Network and spectrum planning are intensive**.
4. **Frequency planning is time-consuming**.
5. **Guard bands waste bandwidth**; bandwidth is also wasted when transmission isn't continuous.

Applications [MA p4, AT]

- **Walkie-talkies** and mobile networks for **closed user groups**.
- **AM/FM radio broadcasting** (each station has its own channel).
- **Early cellular (1G)**, analogue.
- **Cable TV** (coax 4 MHz–1 GHz divided into 6 MHz channels) [AT].
- Fibre optics: **WDM/WDMA** (wavelength division) [AT].
- The old analog telephone hierarchy: 12 voice channels on subcarriers from 60–108 kHz (SSB) [AT].

5.4 Time Division Multiple Access (TDMA) ★

Principle [MA p5–6, MX, AT]

- Digital systems don't need continuous transmission because users don't use their bandwidth all the time.
- Several users share the **same frequency band** by **dividing time into slots** allocated periodically to each user for the duration of a call.
- Possible because TDMA systems **transfer digital data**.
- **Different users can use the same frequency in the same cell, at different times.**
- 2G TDMA systems: **GSM, IS-136, PDC, DECT**.
- The US cellular TDMA standard divides a channel into **six slots, each signal using two**, so **three users per channel**, a **3:1 capacity gain over AMPS**.
- **Registration**: when the user presses push-to-talk (PTT), a control channel registers the radio with the nearest base station, which assigns **a pair of channels** (transmit/receive) **and a time slot**. The user can **only send/receive in its slot**.
- **Information flow is not continuous**: it is sent in **bursts**, re-assembled at the receiver fast enough to sound continuous.
- Analogue calls are **sampled** into digital format and placed into the assigned slot.
- Users need **synchronisation via a global clock** [MX].

Two types [MX]:

- **Fixed TDMA**: slots are statically assigned to data streams, changed only for large traffic variations. Slot size = **T/N** (T = frame time, N = users).
- **Dynamic TDMA**: a **scheduling algorithm** reserves a **variable number of slots** per frame according to traffic demand.



Number of users ★ [MA p6]

$$\text{Users supported} = \text{Number of channels in the spectrum} \times \text{Time slots per channel}$$

Example in the notes' figure: **4 channels × 3 slots = 12 users**.

Worked example TD1 SUPPLEMENTARY (GSM numbers from [AT])

GSM uses 200 kHz channels with 8 slots each. With 25 MHz (one direction) and no guard bands, how many users?

1. Channels = 25 000/200 = **125**
2. Users = 125 × 8 = **1000 simultaneous users** *(With one channel reserved as a guard band, as in real GSM: 124 × 8 = 992.)*

Worked example TD2: combining FDMA and TDMA SUPPLEMENTARY

Bt = 12.5 MHz, Bg = 10 kHz, Bc = 30 kHz, 3 slots/channel (IS-136). N = 416 channels (FD1) → users = 416 × 3 = **1248**.

The T1 and GSM examples [AT]

- **T1**: 24 voice calls; each voice signal (300–3000 Hz) is sampled at **8 kHz** (just above the Nyquist rate) × 8 bits = **64 kbps**. 24 bytes + 1 framing bit = **193 bits per frame**, giving **1.544 Mbps**.
- **GSM**: 200 kHz channels, **8 users per channel**, frame sent at **270 kbps** using **GMSK** (a form of FSK); one frame = 1248 bits.

Advantages of TDMA [MA p7]

1. Data are transmitted in **discrete bursts**, giving:
   - **extended battery life** and talk time compared to FDMA;
   - a **simpler handoff**: the phone can listen to other base stations during idle slots.
2. **More efficient use of spectrum** than FDMA:
   - **accommodates more users** in the same spectrum (capacity in high-traffic metropolitan areas);
   - efficient use of **hierarchical cell structures** (pico, micro, macro).
3. Different slots are used for transmit and receive, so **duplexers are not required**.

Disadvantages of TDMA [MA p7]

1. **Requires synchronisation**: if slot sync is lost, channels may collide.

2. On the uplink, handsets need **high peak power** in transmit mode, which shortens battery life. With n channels the transmit power is **10 log n** dB higher than FDMA.
3. Network and spectrum planning are intensive.
4. **Dropped calls** are possible when users move between cells.
5. **Higher costs** from equipment sophistication.
6. **Equalization is required**, because transmission rates are much higher than in FDMA channels.

5.5 Code Division Multiple Access (CDMA) ★★

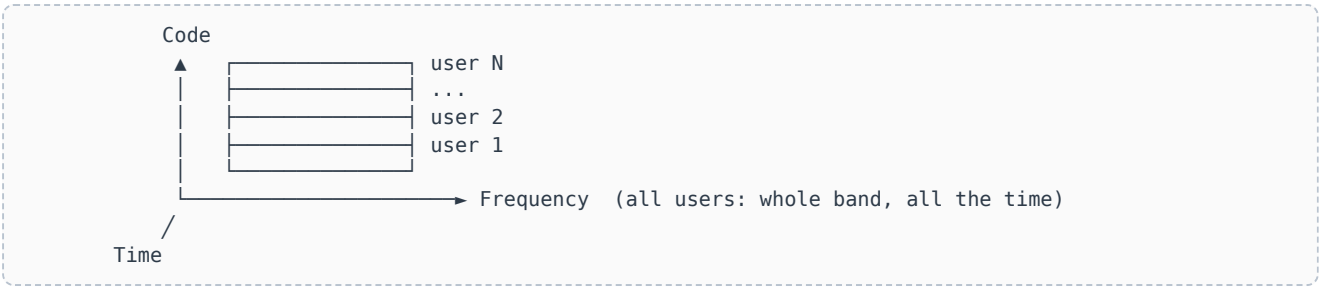
Principle [MA p7–8, MX, AT]

- CDMA allows transmissions to **occupy the entire bandwidth at the same time without interference**.
- Based on **spread spectrum**: an extra modulation expands the bandwidth well beyond what the data need.
- The dominant spread-spectrum methods are **direct-sequence (DSSS)** and **frequency hopping (FHSS)**.
- **Spread spectrum is useful for** [MA p7]:
  1. **Suppressing interference**
  2. **Making interception difficult**
  3. **Accommodating fading and multipath channels**
  4. **Providing a multiple-access capability**
- History: originally **IS-95**, competing with GSM; operates in **800 MHz and 1.9 GHz PCS** bands; developed by **Qualcomm**. Compared to GSM, it needs **fewer towers** and gives **up to 5× the calling capacity**; **>10× the voice traffic of AMPS**; the basis for **3G** [MA p7].
- **Military origin**: first used in **WWII** by the English allies **to foil German attempts at jamming** [MX].

How it works [MA p8]:

1. Each user gets a **unique code sequence** (a **pseudorandom code**) used to code data before transmission; the code is shared by the mobile and the base station.
2. The baseband data are **spread** with this code, which runs at a **much higher rate** (the **chip rate**) than the data.
3. **All users access the entire spectrum all of the time.** The **unique code separates each call** from all others.
4. The transmitted signal sits **below the noise level**.
5. The receiver uses a **correlator** to **despread the wanted signal**, which then passes through a **narrowband filter**.
6. **Unwanted signals are not despread**, so they do not pass the filter; they stay as low-level noise.
7. **Power control**: phones and base stations use the **minimum power** needed, with precise power control to reduce users' power. This also gives **battery life, talk time and smaller batteries** [MA p8].

[AT] adds: data are **XORed with a chipping signal**. IS-95 uses a **1.2288 Mchip/s** chipping signal to spread **13 kbps** voice over **1.25 MHz**. **64 codes = 64 users** in one 1.25 MHz channel. **WCDMA** (3G) uses **3.84 Mchip/s** in **5 MHz**. In [MX], each subscriber is assigned an **orthogonal code called a "chip"**.



Advantages of CDMA [MA p8]

1. **Greatest spectrum efficiency**: capacity about **8–10× analogue** and **4–5× other digital systems**; best in high-traffic areas with limited spectrum.
2. **Improves call quality** by filtering out background noise, crosstalk and interference.
3. **Soft handoffs**: because of multiple diversities, handoffs between cells are **undetected by the user**.
4. **Simplified frequency planning**: all users use the **same frequency spectrum**, so no detailed frequency plans, no re-tunes for expansion, and **fewer cells** for coverage.
5. **Random Walsh codes enhance user privacy**.
6. **Precise power control** increases talk time and battery life.

Disadvantages of CDMA [MA p9]

1. **Backwards-compatibility** techniques are costly.
2. Equipment is **expensive**.
3. **Difficult to optimise** for maximum performance.
4. **Low-traffic areas** make inefficient use of spectrum and equipment.
5. (From the comparison table) **complex receivers**; needs **complicated power control** for senders [MA p10].

Worked example: how CDMA codes separate users SUPPLEMENTARY (illustration of the principle in [MA p8])

Use bipolar values: bit 1 → +1, bit 0 → −1. Two users get **orthogonal (Walsh) codes**:

- User A code: **[+1 +1 +1 +1]**
- User B code: **[+1 −1 +1 −1]**
- Orthogonality check: A·B = (1)(1) + (1)(−1) + (1)(1) + (1)(−1) = **0** ✔

Step 1: spread. A sends bit **1** (+1) → +1 × A = [+1 +1 +1 +1]. B sends bit **0** (−1) → −1 × B = [−1 +1 −1 +1]. **Step 2: the channel adds them.** Received = [0 +2 0 +2]. **Step 3: despread for A.** Received · A = 0 + 2 + 0 + 2 = 4 → divide by 4 chips = **+1 → bit 1 ✔ Step 4: despread for B.** Received · B = 0 − 2 + 0 − 2 = −4 → /4 = **−1 → bit 0 ✔**

Each receiver recovers **only its own data**, even though both signals used the **same frequency at the same time**. A receiver without the right code sees only noise.

Why is CDMA difficult to jam? ★ (Asked in: Exam Q1(d), T1A Q6, T1B Q7)

From the notes:

1. **The signal is spread over a very wide bandwidth** with low power density, **below the noise level** [MA p8]. A narrowband jammer only hits a small part of the band.

2. **Despreading spreads the jammer**: the receiver's correlator multiplies the received signal by the user's code. The wanted signal collapses back to narrowband, while the **jammer (not coded with the user's code) is spread out** and mostly removed by the **narrowband filter** [MA p8].
3. **Pseudorandom code is unknown to the jammer**: without knowing the code (or the hop sequence in FH), a jammer cannot concentrate energy where the signal is. In FHSS, *"a malicious sender may be able to send noise to jam the signal for one hopping period (randomly), but not for the whole period"* [SS].
4. **Designed for this purpose**: spread spectrum's stated uses include **suppressing interference** and **making interception difficult** [MA p7]. It was used in **WWII to foil German jamming** [MX]; Forouzan describes SS as wrapping the message in a "protective envelope" against **jamming from a malicious intruder** [SS].

SUPPLEMENTARY: this resistance is measured by the **processing gain** Gp = Bss/B (= chip rate / bit rate). E.g. IS-95: 1.2288 Mchip/s ÷ 9.6 kbps = 128, i.e. **≈ 21 dB**. A jammer is suppressed by roughly this factor.

5.6 Space Division Multiple Access (SDMA) ★

Principle [MA p9–10, MX, AT]

- Besides frequency, time and code, the **spatial dimension** can separate users: data streams are sent over **different, non-overlapping spatial channels**.
- **SDMA lets users share the same bandwidth simultaneously in different geographical locations.**
- It solves the capacity problem by **exploiting the spatial dimension**: users can be identified **even when they share the same time/frequency/code**.
- Achieved by **beamforming** or **sectorization**.
- **Primitive form**: the **same frequency reused in different cells** (frequency reuse) [MX, AT]. **Sectorized antennas** (three 120° sectors per site) are a primitive form of SDMA [MA p10, AT].
- **Advanced form**: **smart antennas / adaptive arrays** with **dynamic beamforming**, focusing narrow beams on specific users and excluding others [AT]; **spot beams** [MX].
- Areas covered by different beams can use the **same frequency (TDMA or CDMA)** or **different frequencies (FDMA)** [MA p10, MX].
- In practice more than one transmitter may fall inside a beam, so **SDMA is combined with other multiple access techniques** [MX].

Why SDMA was developed: omnidirectional vs adaptive antennas [MA p9–10] | Omnidirectional (traditional) | Adaptive antenna / SDMA | |---|---| | Base station has **no information on the mobile's position**; radiates in **all directions** | Uses **adaptive antenna arrays (smart antennas)** to send/receive **only within a limited angular range**, following the directional nature of multipath | | **Wastes power** where there are no mobiles; causes **interference to co-channel cells** | Improves coverage/link quality when **noise-limited**; **enhances capacity** when **interference-limited** | | Receives noise and interference from all directions | A BS with a **P-element antenna array** serves several single-antenna mobiles at once |

Polarization division (PDMA) [AT]: a variant of SDMA. Two signals on the same frequency use **vertical and horizontal polarization**; the antennas don't respond to the opposite polarization. Widely used in **satellites**.

Advantages of SDMA [MA p10]

1. **Range extension**: fewer cells needed to cover an area.
2. **Interference suppression**: interference from other systems and cells is greatly reduced by exploiting each user's **unique channel impulse response**.
3. **Multipath effect elimination**: multipath copies can be mitigated, or even used to **reinforce** the signal.
4. **Capacity increase**, in two ways:
   - **tighter reuse patterns** (e.g. moving from a **7-cell to a 4-cell pattern nearly doubles capacity**);
   - **intracell reuse**: separate spatial channels on the **same conventional channel within one cell**.
5. **Compatibility** with most existing modulation schemes, carrier frequencies and specifications.

5.7 OFDMA (from the access article) [AT]

- Used in **LTE**. OFDM divides a channel into many narrow **orthogonal** subcarriers (**15 kHz** each) that don't interfere.
- Data are divided into many **lower-speed streams** modulated onto the subcarriers.
- **Each user is assigned a group of subcarriers and time slots.** The smallest group is **12 subcarriers = one resource block (RB)**.
- Very **spectrally efficient** and **less affected by multipath**. (Full detail in Topic 7.)

5.8 Other methods: CSMA [AT]

- **CSMA/CD**: the classical **Ethernet** method. Listen first, transmit if idle; if a collision occurs it is **detected**, and nodes wait a **random time** before retrying.
- **CSMA/CA**: similar, but a scheduling algorithm determines when to transmit. **CSMA/CD is used in wired networks; CSMA/CA is preferred in wireless.** (Full detail in Topic 9.)

5.9 ★ Comparison table [MA p10]

	SDMA	TDMA	FDMA	CDMA
Idea	Segment **space** into cells/sectors	Segment **sending time** into disjoint slots, demand-driven or fixed	Segment the **frequency band** into disjoint sub-bands	**Spread the spectrum** using orthogonal codes
Terminals	Only one terminal active in one cell/sector	All terminals active for short periods on the same frequency	Every terminal has its **own frequency**, uninterrupted	**All terminals active at the same place at the same moment**, uninterrupted
Signal separation	Cell structure, directed antennas	**Synchronization in the time domain**	**Filtering in the frequency domain**	**Code plus special receivers**
Advantages	Very simple; increases capacity per km²	Established, fully digital, very flexible	Simple, established, robust	Flexible, less planning needed, **soft handover**
Disadvantages	Inflexible; antennas typically fixed	**Guard space** needed (multipath); synchronization difficult	Inflexible; frequencies are a scarce resource	**Complex receivers**; needs complicated **power control**

	SDMA	TDMA	FDMA	CDMA
Comment	Only useful combined with TDMA, FDMA or CDMA	Standard in fixed networks; with FDMA/SDMA in many mobile networks	Typically combined with TDMA (frequency hopping) and SDMA (frequency reuse)	Used in many 3G systems; higher complexity; integrated with TDMA/FDMA

5.10 Principles of cellular networks [MX, MA p2]

- Cellular technology is the basis of mobile wireless communication. It supports users where wired networks are hard to provide.
- It was developed to **increase capacity** for mobile radio telephone service, **replacing high-power transmitter/receiver systems**.
- It uses **lower power, shorter range and more transmitters**.
- The area is divided into **cells**, each served by a **base station** (network tower).
- The aim: **increase channel capacity**, i.e. handle as many calls as possible in a given bandwidth with sufficient quality of service.

5.11 Cellular network generations [MA p11–16]

Generation	Key description [MA p11]
0G	Briefcase-size mobile radio telephones
1G	**Analog** cellular telephony
2G	**Digital** cellular telephony
3G	High-speed digital cellular telephony (including **video telephony**)
4G (LTE)	**IP-based** "anytime, anywhere" voice, data and multimedia at faster rates than 3G

Technologies per generation [MA p11]: 1G: **AMPS, NMT, TACS**. 2G: **D-AMPS, GSM/GPRS, cdmaOne**. 3G: **CDMA2000/EV-DO, WCDMA/HSPA+, TD-SCDMA**. 4G: **LTE, LTE-Advanced**. **Services**: 1G analog voice; 2G digital voice + simple data (<0.5 Mbps); 3G mobile broadband (63+ Mbps); 4G faster and better (300+ Mbps; richer content, more connections) [MA p11]. **Evolution summary** [MA p12]: 1G established **seamless mobile voice**; 2G **increased voice capacity** ("mobile to the masses"); 3G **optimised mobile for data** (mobile broadband); 4G LTE delivers **more capacity** for faster broadband.

Details per generation [MA p13–15]

1G: circuit-switched; **FDMA**; analog; basic mobility; **poor voice quality**; **poor security**. Frequency **150 / 900 MHz**; analog bandwidth (**30 kHz**); data rate **2 kbps**; **1980–1990**; poor battery, big phones; "better than nothing".

2G: frequency **1.8 GHz (900 MHz)**; bandwidth 900 MHz (25 MHz); **digital**; **GSM**; **64 kbps**; **1991–2000**; **text messaging**; the signal must be strong or the digital signal is weak. **2.5G**: 2G with **GPRS**: e-mail, web browsing, camera phones.

3G: frequency **1.6–2.0 GHz**; bandwidth **100 MHz**; digital broadband with increased speed; **CDMA, UMTS, EDGE**; **144 kbps – 2 Mbps**; **2000–2010**; smartphones, **video calls**, mobile TV; phones expensive.

4G: frequency **2–8 GHz**; bandwidth 100 MHz; high speed, **all-IP**; **LTE, WiFi**; **100 Mbps – 1 Gbps**; **2010–(2020)**; **MAGIC** = **M**obile multimedia, **A**nytime anywhere, **G**lobal mobile support, **I**ntegrated wireless

solutions, **C**ustomized personal service; good QoS + high security; **bigger battery usage**.

5G: **2020–2030**; capacity **1 Gbps – "unlimited"**; high speed and capacity; faster than 4G; supports interactive multimedia, voice streaming; more efficient; frequency **3–300 GHz**; **1000× bandwidth per unit area**; technology **WWWW** (World Wide Wireless Web).

Comparison table [MA p16]

	1G	2G	3G	4G	5G
Deployment	1970–80	1990–2004	2004–10	Now	Soon (≈2020)
Data bandwidth	2 kbps	64 kbps	2 Mbps	1 Gbps	> 1 Gbps
Technology	Analog	Digital	CDMA 2000, UMTS, EDGE	Wi-Max, Wi-Fi, LTE	WWWW
Core network	PSTN	PSTN	Packet network	Internet	Internet
Multiplexing	**FDMA**	**TDMA/CDMA**	**CDMA**	**CDMA**	**CDMA**
Switching	Circuit	Circuit, packet	Packet	All packet	All packet
Primary service	Analog phone calls	Digital calls + messaging	Calls, messaging, data	All-IP service (incl. voice)	High speed, high capacity, Gbps broadcasting
Key differentiator	Mobility	Secure, mass adoption	Better Internet experience	Faster broadband, lower latency	Better coverage, no dropped calls, much lower latency
Weakness	Poor spectral efficiency, major security issue	Limited data rates; hard to support Internet/e-mail	Real performance fails to match hype; failure of WAP	More battery use; complicated, expensive hardware	?

> **NOTES ERROR**: the multiplexing row lists **CDMA for 4G and 5G**. That is inaccurate. **4G LTE uses OFDMA** (downlink) and **SC-FDMA** (uplink), as your own OFDM notes state [OF §8.2–8.3] and the article confirms [AT]. **5G NR also uses OFDM-based access.** If asked about 4G multiple access, answer **OFDMA**. Also note that the two "period" tables in your notes disagree (1G "1980–1990" vs "1970–80"). Quote the period with "approximately".

5.12 Examination focus — Topic 5

- Define multiple access; list the four techniques.
- FDMA/TDMA/CDMA/SDMA: principle, advantages, disadvantages, applications (4–10 marks each).
- **Describe CDMA and explain why it is difficult to jam** (4–10 marks; asked **3 times**).
- Compare FDMA, TDMA, CDMA and SDMA (the comparison table).
- Calculations: FDMA channels N = (Bt − 2Bg)/Bc; TDMA users = channels × slots.
- 1G–5G comparison; MAGIC.

5.13 Topic 5 revision summary

- **FDMA**: frequency bands + guard bands; one user per channel, 100% of the time; simple filters; wastes idle channels and guard bands. **N = (Bt − 2Bg)/Bc.**
- **TDMA**: one frequency, time slots, bursts; needs **sync** and **equalization**; better spectrum use, battery life, simple handoff; no duplexer. **Users = channels × slots.**
- **CDMA**: whole band, all the time, **unique pseudorandom codes**, chip rate ≫ bit rate, correlator despreads; **power control**; soft handoff; 4–5× capacity of other digital systems; **hard to jam/intercept**.
- **SDMA**: space; beamforming, sectorization, frequency reuse; range extension, interference suppression, capacity.
- Generations: 1G analog FDMA → 2G digital GSM TDMA → 3G CDMA/UMTS → 4G all-IP LTE (OFDMA) → 5G.

TOPIC 6 — SPREAD SPECTRUM

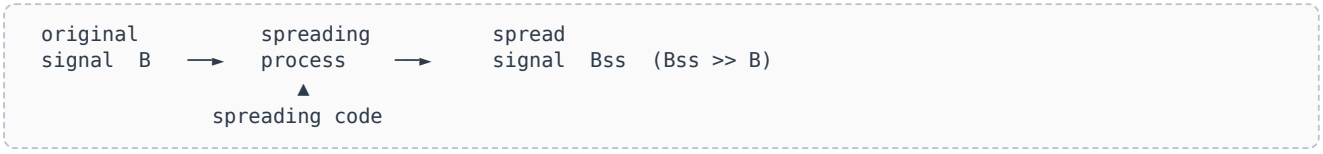
6.1 Introduction

Multiplexing combines signals to achieve **bandwidth efficiency**. **Spread spectrum** also combines signals into a larger bandwidth, but with **different goals**: in wireless, all stations share the air, so they must share the medium **without interception by an eavesdropper** and **without being jammed by a malicious intruder** (e.g. military) [SS].

To achieve this, spread spectrum **adds redundancy**: the bandwidth B each station needs is **expanded to Bss, where Bss ≫ B**. Forouzan's analogy: like putting a delicate, expensive gift in a **protective box** and using a superior delivery service [SS].

6.2 Two principles of spread spectrum ★ [SS]

1. The bandwidth allocated to each station must be **far larger than what is needed**. This allows **redundancy**.
2. The expansion from B to Bss must be done **by a process independent of the original signal**: the spreading happens **after** the source creates the signal, using a **spreading code** (a series of numbers that **look random but are actually a pattern**).



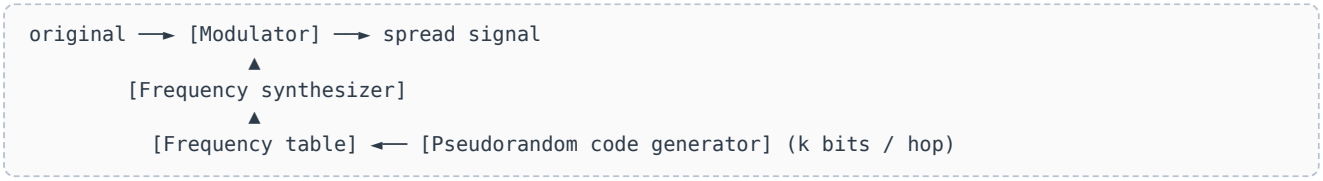
Two techniques: **FHSS** and **DSSS** [SS].

Wireless benefits (also from [WC-B 5.4]): spread spectrum uses **100× or more bandwidth** with **lower energy density** per frequency. Even if some frequencies are poor (frequency-selective fading), average performance is good. It is a form of **frequency diversity** [WC-S 48].

6.3 Frequency Hopping Spread Spectrum (FHSS) ★

How it works [SS Fig 6.28]

- Uses **M different carrier frequencies** modulated by the source signal. At any moment the signal modulates **one** carrier; at the next moment, **another**. Over time all M are used, so **B_FHSS ≫ B**.
- Components:
  1. A **pseudorandom code generator (PN)** creates a **k-bit pattern every hopping period Th**.
  2. A **frequency table** maps the pattern to a frequency.
  3. A **frequency synthesizer** creates that carrier.
  4. The **modulator**: the source signal modulates the carrier.



The notes' worked example [SS Fig 6.29–6.30]

- **M = 8** hopping frequencies, so **k = 3** bits ($2^3 = 8$). The original signal needs **100 kHz**.
- Frequency table: 000→200 kHz, 001→300, 010→400, 011→500, 100→600, **101→700**, 110→800, **111→900 kHz**.
- PN pattern: **101, 111, 001, 000, 010, 110, 011, 100**, then it **repeats after 8 hops**.
- Hop 1: 101 → **700 kHz**; hop 2: 111 → **900 kHz**; ...; hop 8: 100 → **600 kHz**; then back to 101.

Hop	1	2	3	4	5	6	7	8
Pattern	101	111	001	000	010	110	011	100
Carrier (kHz)	700	900	300	200	400	800	500	600

> **NOTES ERROR (minor)**: the text in [SS] lists the pattern as "101, 111, 001, 000, 010, 011, 100" (only 7 patterns, missing **110**). The figure shows all eight: 101 111 001 000 010 **110** 011 100. Use the eight-pattern version.

Why FHSS achieves the goals [SS]

- **Privacy**: with many k-bit patterns and a short hopping period, an intruder can only catch a **small piece** of data, because they don't know the sequence and can't follow the hops.
- **Anti-jamming**: a jammer may jam **one hopping period** (randomly), **not the whole transmission**.

Bandwidth sharing [SS Fig 6.31]

- With **M hopping frequencies**, **M channels can be multiplexed** into the same Bss: each station uses one frequency per hop, leaving M − 1 for M − 1 other stations (with a suitable modulation such as MFSK).
- **FHSS vs FDM**: in **FDM** each station uses **1/M of the band, fixed**. In **FHSS** each station also uses 1/M, but **the allocation changes from hop to hop**.

FHSS in the module's technologies

- **IEEE 802.11 FHSS**: 2.4 GHz ISM; 2- or 4-level FSK; 1 or 2 Mbps; band divided into **79 sub-bands of 1 MHz**; a pseudorandom generator selects the hop sequence [WLAN slides 23–24].
- **Bluetooth**: **79 × 1 MHz channels**, **1600 hops/s** (slot 0.625 ms); piconet access **FH-TDD-TDMA**; scatternet access **FH-CDMA** [BT].

6.4 Direct Sequence Spread Spectrum (DSSS) ★

How it works [SS Fig 6.32]

- Also expands bandwidth, but differently: **each data bit is replaced by n bits (chips)** using a spreading code. The **chip rate is n times the data bit rate**.
- A **chips generator** feeds a **modulator** together with the original signal.

The notes' example: the Barker sequence [SS Fig 6.33]

- Used in **wireless LANs**: the **Barker sequence, n = 11**: **10110111000**.

- With polar NRZ, **spread signal = original × chips**: a data bit 1 sends the code as is; a data bit 0 sends the **inverted** code.
- If the original rate is N, the spread rate is **11N**, so the spread signal needs **11 times the bandwidth**.

Worked example DS1: spreading data 1 0 1 with the Barker code | Data bit | Chips sent (11 chips per bit) | |---|---| | 1 | 1 0 1 1 0 1 1 1 0 0 0 (code as is) | | 0 | 0 1 0 0 1 0 0 0 1 1 1 (code inverted) | | 1 | 1 0 1 1 0 1 1 1 0 0 0 |

SUPPLEMENTARY: in binary (0/1) form, spreading = **data XOR chip** with the convention "data 1 ↔ chip unchanged". Using bipolar values (+1/−1) it is plain multiplication. At the receiver, multiplying by the same code again recovers the data (correlation).

- **Benefits**: **privacy** if the intruder doesn't know the code; **immunity against interference** if each station uses a different code [SS].
- **Bandwidth sharing in DSSS: "no and yes"** [SS]: if the code's spread signals **cannot be combined and separated**, the bandwidth **cannot** be shared (e.g. some WLANs using the Barker code). With **special codes that allow combining and separating** (orthogonal codes, i.e. **CDMA**), it **can** be shared.

DSSS in the module's technologies

- **802.11 DSSS**: 2.4 GHz ISM; **BPSK/QPSK**; 1 or 2 Mbps. **802.11b HR-DSSS**: **5.5 and 11 Mbps** [WLAN slides 24–25].
- **CDMA (IS-95)**: 1.2288 Mchip/s, 1.25 MHz channel [AT].
- **802.15.4/ZigBee**: DSSS PHY at 868/915 MHz and 2.4 GHz [BT].

6.5 ★ FHSS vs DSSS comparison (assembled from [SS], [MA p7])

	FHSS	DSSS
Spreading method	Carrier **hops** among M frequencies in a pseudorandom sequence	Each bit is replaced by **n chips** (code at n× the bit rate)
Bandwidth at any instant	**Narrow** (one channel)	**Wide** (whole spread band all the time)
Code determines	**Which frequency** to use in each hop period	**The chip pattern** that multiplies each bit
Key components	PN generator, frequency table, frequency synthesizer, modulator	Chip generator, modulator (XOR/multiply)
Effect of narrowband interference/jamming	Only the hops that land on the jammed frequency are hit	The receiver's despreading **spreads the jammer**, which is filtered out
Sharing	M stations can share Bss (like time-varying FDM)	Shareable only with orthogonal codes (CDMA)
Examples	802.11 FHSS, **Bluetooth**	802.11 DSSS/b, **IS-95 CDMA**, WCDMA, 802.15.4

FH-CDMA vs DS-CDMA (Bluetooth exercise 15): in **FH-CDMA**, users are separated because they follow **different hopping sequences**; they occasionally collide on the same frequency (e.g. Bluetooth piconets in a scatternet [BT]). In **DS-CDMA**, users transmit **at the same time over the same wide band**, separated by **different (orthogonal/pseudorandom) chip codes** and correlation at the receiver (e.g. IS-95) [MA p7–8, AT].

6.6 Examination focus — Topic 6

- **Explain the spread spectrum principles of (i) FHSS and (ii) DSSS** (asked in **Exam Q1(b), T1A Q4, T1B Q6**; 4–10 marks).
- **Describe CDMA and why it is difficult to jam** (3 times).
- Spread spectrum vs multiplexing; two principles of SS; the Barker code; the FHSS example table; FH-CDMA vs DS-CDMA.

6.7 Topic 6 revision summary

- SS goals: **privacy and anti-jamming**, not bandwidth efficiency. **Bss ≫ B**, added **redundancy**; spreading is **independent of the signal**, using a **pseudorandom code**.
- **FHSS**: PN generator → frequency table → synthesizer → modulator; hops every Th; M frequencies, k = $\log_2 M$ bits. Jammer hits only some hops.
- **DSSS**: each bit → n chips; chip rate = n × bit rate; **Barker 11 chips = 10110111000**; bandwidth × n.
- **CDMA** = DS spread spectrum with orthogonal codes, so many users share the band.

TOPIC 7 — OFDM, OFDMA AND SC-FDMA

7.1 Introduction

OFDM has *"created great expansion in the capacity of wireless networks"*. It is the main air-interface technology used to move from **3G to 4G**, allows the expansion of **IEEE 802.11** data rates, and is critical to **WiMAX** broadband [OF]. Together with **MIMO**, OFDM is the **cornerstone of broadband wireless** [WC-B 5.4].

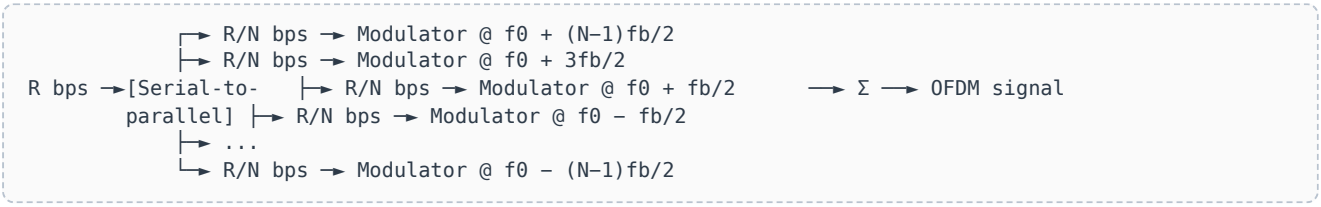
Learning objectives [OF]: overview of OFDM, OFDMA and SC-FDMA; how OFDM **combats frequency-selective fading**; the value of **orthogonal carriers**; the reduction in oscillators using the **IFFT**; problems of **PAPR**; how multiple access is achieved using OFDM and SC-FDMA.

7.2 Key definitions

Term	Definition	Source
OFDM (multicarrier modulation)	Uses **multiple carrier signals at different frequencies**, sending some of the bits on each. Similar to FDM, **but many subcarriers are dedicated to a single data source**	[OF §8.1]
Orthogonality	Two signals $s_1(t)$, $s_2(t)$ are orthogonal if the **average over a bit time of $s_1(t) \cdot s_2(t) = 0$**	[OF §8.1]
OFDMA	Combination of **FDMA and TDMA**: different users use **subsets of the subcarriers** at **different times**	[OF §8.2]
Subchannel	A **group of subcarriers** (scheduling individual subcarriers is not feasible or sufficient); called a **Resource Block** in LTE	[OF §8.2]
SC-FDMA	Single-carrier FDMA: similar structure and performance to OFDMA, but with **lower PAPR**; performs an extra **DFT** before the IFFT	[OF §8.3]

7.3 How OFDM works ★ [OF §8.1, Fig 8.1]

- A binary stream at **R bps**, with available bandwidth **N·fb** centred on f_o.
- **Option 1**: use the whole bandwidth for one stream, so each bit lasts **1/R**.
- **OFDM**: a **serial-to-parallel converter** splits the stream into **N substreams**, each at **R/N bps**, each on its own **subcarrier**, spaced **fb** apart. Now each bit lasts **N/R**, which is **much longer**. This gives special capabilities to **overcome multipath fading**.



Orthogonality ★ [OF §8.1, Fig 8.2]

- The **base frequency fb** is the lowest subcarrier; all others are **integer multiples**: 2fb, 3fb, ...
- The spectra **overlap substantially**, but **the peak of each subcarrier's spectrum falls where every other subcarrier's power is zero**, so there is **no interference** between them.

- **Traditional FDM** spaces carriers apart to (1) avoid overlap and (2) add **guard bands**. In the notes' comparison (Fig 8.2b vs 8.2c, subcarriers at fb, 2fb, 3fb vs fb, 7fb, 13fb), **OFDM supports 6 times as many signals** in the same spectrum.

Mathematical condition [OF]:

$$\text{Average over bit time of } s_1(t)\cdot s_2(t) = 0$$

At the transmitter $s(t) = s_1(t) + s_2(t)$. To extract s_1, the receiver multiplies by $s_1(t)$ and averages:

$$\text{avg}[s_1(t)\cdot s(t)] = \text{avg}[s_1^2(t)] + \text{avg}[s_1(t)\cdot s_2(t)] = \text{avg}[s_1^2(t)] + 0$$

Only s_1 remains; all other orthogonal signals are removed.

Requirement for OFDM subcarriers: if the subcarrier bit time is **T**, then **fb must be a multiple of 1/T**, and every other subcarrier is M·fb = M/T for an integer M [OF]. **Examples**: **LTE subcarrier spacing = 15 kHz**. **802.11n**: **48 subcarriers in 20 MHz** (108 in 40 MHz), **fb = 0.3125 MHz**, then translated to 2.4 or 5 GHz [OF].

Benefits of OFDM ★ [OF §8.1]

1. **Frequency-selective fading only affects some subcarriers**, not the whole signal. With **FEC**, this is easily handled.
2. **Overcomes intersymbol interference (ISI)** (the most important benefit). The data rate per carrier drops by **N**, so the **symbol time grows by N**: from **Ts to N·Ts**. Choose N so that **N·Ts ≫ the RMS delay spread** of the channel, and multipath delays hardly matter.
3. As a result **equalizers may not be needed**. Equalizers are complex and expensive, and their complexity grows with ISI severity.
4. **Spectral efficiency**: orthogonal subcarriers pack tightly, with no guard bands between them (6× in the example).
5. [WLAN slide notes] **Degradations are easier to handle per subcarrier**: badly degraded subcarriers can be **excluded** in favour of good ones.

Difficulties of OFDM SUPPLEMENTARY (pages 241–244 of [OF] are missing from your upload; the learning objectives name these topics)

1. **Peak-to-average power ratio (PAPR)**: adding many subcarriers occasionally produces **very high peaks**, so the transmitter's power amplifier must be **linear over a wide range**. This is expensive and **power-inefficient**, which is bad for battery-powered phones. This is why LTE uses **SC-FDMA on the uplink**.
2. **Intercarrier interference (ICI)**: orthogonality depends on exact spacing; **frequency offsets and Doppler** break it, so subcarriers leak into each other. Tight synchronisation is needed.
3. **IFFT implementation**: building N separate oscillators would be impractical. The **inverse fast Fourier transform (IFFT)** generates all subcarriers digitally at once (and the FFT at the receiver), which is why OFDM is practical. A **cyclic prefix** (copy of the end of each symbol placed at its start) acts as a guard interval against ISI [OF Fig 8.8 shows "Add CP"].

7.4 OFDMA ★ [OF §8.2]

Principle

- Multiple access options: scheduled times (**TDMA**), random access (**CSMA**), scheduled frequencies (**FDMA**), coded spreading (**DSSS**), coded hopping (**FHSS**).
- **OFDMA = FDMA + TDMA**: users get **subsets of subcarriers at different times**.
- **Not all OFDM systems use OFDMA**: e.g. some 802.11 versions use OFDM for transmission but **CSMA** for access, and the transmitter uses **all** subcarriers when sending. **LTE uses OFDMA only on the downlink** and a **single-carrier approach (SC-FDMA) on the uplink**.
- Subcarriers are grouped into **subchannels**. In the **downlink**, subchannels may go to **different receivers**. In the **uplink**, a transmitter may be assigned one or more subchannels.
- **Uplink subchannelization saves transmit power**: the device concentrates its power on its own subchannels, which is useful for **battery-powered 4G devices**.

★ OFDM vs OFDMA (Asked in: T1B Q8, 10 marks; OF review Q 8.5)

	OFDM	OFDMA
What it is	A **modulation / transmission technique** (multicarrier)	A **multiple access technique** built on OFDM
Users per symbol	**All subcarriers carry one user's data** at a time	Subcarriers are **divided among several users** at the same time
Sharing between users	Needs another method, e.g. **TDMA or CSMA** (802.11a/g)	Built in: **FDMA + TDMA** (subchannels × time slots)
Allocation unit	Whole channel	**Subchannel / resource block** (LTE: 12 subcarriers)
Scheduling	None per subcarrier	**Opportunistic scheduling** by channel quality
Power (uplink)	Device spreads power over all subcarriers	Device concentrates power on **its subchannels**
Examples	802.11a/g/n	**LTE downlink**, WiMAX

7.5 ★ The three subchannel allocation methods (Asked in: Exam Q1(c), T1A Q5; OF review Q 8.10)

Method	Description	**Benefits**	Drawbacks	Used by
(i) Adjacent subcarriers	All subcarriers of a subchannel in a **contiguous block**	All **SINRs ≈ equal**. Gives an **opportunity**: the system can choose among many blocks the **best block for each user**, optimising **channel efficiency, user requirements, priority and fairness** (exploits frequency-selective fading for multi-user gain)	If those frequencies are poor, the whole subchannel suffers. Requires accurate knowledge of each user's channel over the **full bandwidth** → **pilot symbols**, prediction (information may already be **out of date**)	LTE, WiMAX

Method	Description	**Benefits**	Drawbacks	Used by
(ii) Regularly spaced subcarriers ("distributed" / "comb")	Subcarriers spread at **regular intervals** across the band	**Frequency diversity**: there are enough **good subcarriers regardless of which are chosen**; **channel estimation burden is much lower**	Can't exploit the single best block for a user	LTE
(iii) Randomly spaced subcarriers	Subcarriers chosen **randomly** across the band	**Same frequency-diversity benefits** as regularly spaced, **plus reduced adjacent-cell interference** (neighbouring cells' random patterns rarely collide fully)		WiMAX

7.6 Opportunistic scheduling [OF §8.2]

Subchannels (**Resource Blocks** in LTE) are allocated to subscriber stations by **channel conditions and data requirements**, possibly with prescribed power levels. Time-varying, multipath channels create **opportunities** that can be exploited. Most standards **don't prescribe** the scheduler (vendors compete). Considerations:

1. **Efficiency**: schedule users with the **highest SINR**; they use AMC for high throughput, which maximises total capacity.
2. **Fairness**: pure efficiency starves users far from the base station. **Proportional fairness**: each user computes $\mathbf{r_i / \bar{r}_i}$ (rate achievable now ÷ average rate obtained). Users are compared **against themselves**; those with a good opportunity **for them** get scheduled.
3. **Requirements**: delay and jitter for audio and video.
4. **Priority**: police, fire, ambulance and public-safety users in emergencies.

Worked example PF1 SUPPLEMENTARY (applying the notes' formula)

User 1: achievable now r_1 = 6 Mbps, average $\bar{r}_1$ = 4 Mbps → metric **1.5**. User 2 (cell edge): r_2 = 2 Mbps, average $\bar{r}_2$ = 1 Mbps → metric **2.0**. **User 2 is scheduled**, even though user 1's absolute rate is higher, because user 2 currently has an unusually good channel *for them*.

7.7 SC-FDMA [OF §8.3]

- Similar structure and performance to OFDMA. Its prominent advantage is a **lower PAPR**, which gives **battery life, power efficiency and lower cost** for mobiles.
- Not the same as basic OFDM: it adds an **extra DFT** at the transmitter and **frequency equalization** at the receiver.
- Used on the **uplink** (LTE); **OFDMA stays on the downlink** for greater multiple-access flexibility.
- **Chain** (Fig 8.8): bit stream → **N-point DFT** → subcarrier mapping → **Q-point IFFT** → add **CP** → channel → remove CP → FFT → de-mapping → **EQ** → **IDFT** → bits. (OFDMA omits the DFT/IDFT.)
- The DFT **spreads each data symbol over all subcarriers**, giving a **virtual single-carrier** structure. **Every data symbol is carried by every subcarrier.**
- **Multiple access**: in OFDMA, different users can share subcarriers **in the same interval**. In SC-FDMA, at any instant **all subcarriers carry the same stream, so they must belong to one user**; multiple access is only **over time**. A better name would be "SC-OFDM-TDMA".

7.8 Examination focus — Topic 7

- Benefits of the three OFDMA approaches (**Exam Q1(c) 6, T1A Q5 9**).
- Difference between OFDM and OFDMA (**T1B Q8 10**).
- Review questions [OF]: define OFDM, OFDMA, SC-FDMA; the **fundamental requirement of OFDM** (**orthogonality**: subcarrier spacing a multiple of 1/T); main strengths; main problems (**PAPR, ICI**); how orthogonality increases capacity (overlapping without interference, ~6×); the purpose of SC-FDMA (**lower PAPR on the uplink**).

7.9 Topic 7 revision summary

- OFDM: split R bps into **N streams of R/N** on **orthogonal subcarriers** (multiples of fb; spacing = 1/T). Symbol time N·Ts ≫ delay spread → **no ISI**, often no equalizer. Selective fading hits only a few subcarriers (FEC fixes them). Tight packing.
- LTE 15 kHz spacing; 802.11n 48 subcarriers/20 MHz, fb = 312.5 kHz.
- OFDMA = FDMA + TDMA over **subchannels/resource blocks**; LTE **downlink**. Allocation: **adjacent** (best-block selection, needs full CSI), **regular** (diversity, low estimation burden), **random** (diversity + less adjacent-cell interference).
- Scheduling: efficiency, **proportional fairness $r_i/\bar{r}_i$**, requirements, priority.
- SC-FDMA: extra DFT, **low PAPR**, LTE **uplink**; one user at a time.

TOPIC 8 — ERROR DETECTION AND CORRECTION

8.1 Introduction

Wireless channels have high bit error rates (Topic 2). Receivers must at least **detect** errors (then use ARQ retransmission) or, better, **correct** them (FEC). The worksheet [ED] lists the four methods examined in this module: **parity checks, the Internet checksum, the Hamming code and the cyclic redundancy check (CRC)**. The calculation questions on these appear in **every** Test 1/Test 2/Exam paper.

> **Source note**: the worksheet [ED] contains web links and worked CRC examples, but **no written theory** for parity, checksum or Hamming. Theory below marked SUPPLEMENTARY follows the standard sources the worksheet links to (UNB ECE4253 pages; Kurose & Ross; Forouzan). The **error detection process** and FEC/ARQ come from your notes [WC-S 69–71; WC-B 5.4].

8.2 Key definitions

Term	Definition	Source
Error detection code	Detects the **presence** of an error	[WC-S 69]
ARQ (automatic repeat request)	The block with an error is **discarded** and the transmitter **retransmits** it	[WC-S 69]
Error correction code (FEC)	Designed to **detect and correct** errors	[WC-S 69]
Single-bit error	**Only one bit** of the data unit (byte, frame, packet) is changed (1→0 or 0→1)	SUPPLEMENTARY (Forouzan)
Burst error	**Two or more bits** in the data unit have changed (not necessarily consecutive)	SUPPLEMENTARY (Forouzan)
Burst length	Measured **from the first corrupted bit to the last corrupted bit**, inclusive; bits in between may or may not be corrupted	SUPPLEMENTARY (Forouzan)
Codeword	Data bits plus the redundant (check) bits actually transmitted	SUPPLEMENTARY
Modulo-2 arithmetic	Binary addition/subtraction **without carries or borrows**, identical to **XOR**	SUPPLEMENTARY

8.3 The error detection process [WC-S 70–71]

Transmitter: for a given frame, an error-detecting code (**check bits**) is calculated from the data bits and **appended** to them. **Receiver**: separates the frame into data and check bits, **recalculates** the check bits from the received data, **compares** them with the received check bits. A **mismatch = detected error**.

```
Tx:  data ──► f(data) = E ──► send [data | E]
Rx:  receive [data' | E'] ──► compute f(data') = E'' ──► compare E'' with E'
                                                          equal → accept
                                                          differ → error detected (ARQ / discard)
```

8.4 ★ Single-bit errors vs burst errors (Asked in: T1B Q1 [10], T2-2026 Q5

[3])

SUPPLEMENTARY theory (standard Forouzan/Stallings definitions):

	Single-bit error	Burst error
Definition	**Only 1 bit** in the data unit is changed	**2 or more bits** changed
Typical cause	Short, isolated noise; rare in serial transmission	**Impulse noise**, fading dips. A noise burst lasts longer than one bit, so it hits many bits. In wireless, **burst errors dominate**
Example	Sent 0000 0010 → received 0000 **1**010	Sent 0100 0100 0100 0011 → received 0101 1101 0100 0011 (errors at positions 4, 5 and 8 → burst length 8 − 4 + 1 = 5)
Length	1 bit	From the **first to the last corrupted bit** inclusive

Your notes support the cause: **impulse noise is the primary source of error in digital data**. A 0.01 s spike wipes out **~10 000 bits at 1 Mbps**, which is a long burst [WC-B 5.2]. Fading in mobile channels causes 20–30 dB drops lasting many bits [WC-B 5.3].

Measuring burst length (worked): sent 1011 0110, received 10**0**1 0**0**10. Errors at bit positions 3 and 6 (from the left). Burst length = 6 − 3 + 1 = **4 bits** (positions 4 and 5 are inside the burst but happen to be correct).

8.5 ★ Parity checking

SUPPLEMENTARY theory (standard; the worksheet's first link covers it).

Single (even) parity: append **one bit** so the total number of 1s (data + parity) is **even**. (Odd parity: make it odd.)

- Data 1011001 (four 1s) → even parity bit **0** → send 1011001**0**.
- Data 1011000 (three 1s) → even parity bit **1** → send 1011000**1**.
- **Receiver**: counts the 1s. Odd count → **error detected**.

★ What parity can and cannot do (Asked in: T1B Q1 [10], T2-2026 Q6 [5])

1. **Single-bit errors: always detected** ✔. One flip changes the count of 1s by one, so even becomes odd.
2. **Burst errors: NOT guaranteed.** Parity detects **any odd number** of flipped bits but **misses any even number** (2, 4, ...), because two flips leave the count's parity unchanged.
   - Example: send 1011001**0** (four 1s). A burst flips bits 2 and 3 → 1**10**10010 still has four 1s → **passes the check** → undetected.
   - A burst flips 3 bits → odd count → detected. **About half of random bursts escape detection.**
3. **Correction: impossible** with one parity bit. The receiver knows *that* an odd number of bits changed, but **not which bit**; any of the n bits could be the culprit. Only ARQ (retransmission) can recover.

Conclusion on appropriateness: parity is **appropriate for single-bit errors** (cheap, 1 extra bit, always detects them) but **inappropriate for burst errors**, which are the common case in wireless/noisy channels. For bursts, use **CRC** (which detects all bursts of length ≤ r, where r = number of check bits) or **two-dimensional parity**.

Two-dimensional parity SUPPLEMENTARY: arrange data in rows and add a parity bit to **every row and every column**. It can **detect and correct any single-bit error** (the failing row and column intersect at the

bad bit) and detects many burst errors, but some patterns of **4 errors in a rectangle** go undetected.

```
           data        row parity
         1 0 1 1   |   1
         0 1 1 0   |   0
         1 1 0 1   |   1
         ----------
col par  0 0 0 0   |   0
A single error at row 2, col 3 makes row 2 AND column 3 fail → bit located → flip it.
```

8.6 ★ The Internet checksum

SUPPLEMENTARY theory (standard RFC 1071 / Kurose method, at the 4-bit word size used in your questions).

Sender:

1. Divide the message into **k-bit words** (here k = 4).
2. Add all words using **one's-complement addition**: any **carry out of the top bit is wrapped around** and added to the result (**end-around carry**).
3. The **checksum = one's complement of the sum** (flip every bit).
4. Send the data words + the checksum.

Receiver: adds **all words including the checksum** with end-around carry. If there are no errors, the result is **all 1s (1111)**, and its complement is 0000. Anything else → error.

★ Worked example CK1 (Exam Q2(c), 2 marks; also the worksheet [ED])

Message 1001 1100 1010 0011, Internet checksum with 4-bit words. Value of the checksum?

Step	Operation	Binary result	Carry?	After wrap
1	1001 + 1100	1 0101	carry 1	0101 + 1 = **0110**
2	0110 + 1010	1 0000	carry 1	0000 + 1 = **0001**
3	0001 + 0011	0100	none	**0100**

- **Sum = 0100**
- **Checksum = one's complement of 0100 = 1011**

Check (decimal): 9 + 12 + 10 + 3 = 34. One's-complement 4-bit sum: 34 = 2 × 15 + 4 → 4 = 0100 ✔ → complement **1011**. **Receiver check**: 0100 (sum of data) + 1011 = **1111** ✔ (no error).

★ Worked example CK2 (T2-2026 Q4: (a) 3 marks, (b) 1 mark)

Message 1010 1100 0110 1001, 4-bit words, one's-complement addition with end-around carry.

Step	Operation	Binary result	Carry?	After wrap
1	1010 + 1100	1 0110	carry 1	0110 + 1 = **0111**
2	0111 + 0110	1101	none	**1101**
3	1101 + 1001	1 0110	carry 1	0110 + 1 = **0111**

- **Sum = 0111 → Checksum = 1000**
- **Check (decimal)**: 10 + 12 + 6 + 9 = 37 = 2 × 15 + 7 → 0111 ✔.

- **(b)** The receiver adds all message words **and** the checksum: 0111 + 1000 = **1111**. So the expected result is **1111** (all ones; its complement is 0000).

8.7 ★ Hamming code

SUPPLEMENTARY theory (the worksheet links the UNB Hamming page; **this is the method T2-2026 Q2 specifies**).

Purpose: an **error-correcting code** (FEC) that can **detect and correct any single-bit error** by using several parity bits, each covering a different subset of positions. The pattern of failing checks **points to the bad bit**.

Step 1: how many check bits? The Hamming inequality

For m data bits and r check bits:

$$2^r \geq m + r + 1$$

Why: the r check bits can give 2^r different "syndromes". These must name every one of the **m + r positions** where a single error could be, **plus 1** for "no error".

For **m = 8**:

- r = 3: $2^3 = 8 \geq 8 + 3 + 1 = 12$? **No ✘**
- r = 4: $2^4 = 16 \geq 8 + 4 + 1 = 13$? **Yes ✔**
- **Minimum r = 4 check bits**, so the codeword has **12 bits**.

Step 2: positions

- Number positions **1 to 12 from the left**.
- **Check bits go at positions that are powers of 2: 1, 2, 4, 8.**
- **Data bits fill the remaining positions 3, 5, 6, 7, 9, 10, 11, 12, left to right.**

Step 3: which positions each check bit covers

Check bit P at position 2^k covers every position whose **binary number has that bit set**:

Check bit	Covers positions (rule)	Data positions it checks
P1 (pos 1)	binary xxx**1**: 1, 3, 5, 7, 9, 11	**3, 5, 7, 9, 11**
P2 (pos 2)	binary xx**1**x: 2, 3, 6, 7, 10, 11	**3, 6, 7, 10, 11**
P4 (pos 4)	binary x**1**xx: 4, 5, 6, 7, 12	**5, 6, 7, 12**
P8 (pos 8)	binary **1**xxx: 8, 9, 10, 11, 12	**9, 10, 11, 12**

Even parity: each check bit = XOR (sum mod 2) of the data bits it covers, so the group's total number of 1s is even.

★ Worked example HM1 (Exam Q2(a), 4 marks; also the worksheet [ED]): data 10101111

Check bits needed: m = 8 → **r = 4** (inequality above).

Placement: | Position | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | |---|---|---|

Check bits (even parity):

- **P1** = bits at 3, 5, 7, 9, 11 = 1, 0, 0, 1, 1 → three 1s → **P1 = 1**
- **P2** = bits at 3, 6, 7, 10, 11 = 1, 1, 0, 1, 1 → four 1s → **P2 = 0**
- **P4** = bits at 5, 6, 7, 12 = 0, 1, 0, 1 → two 1s → **P4 = 0**
- **P8** = bits at 9, 10, 11, 12 = 1, 1, 1, 1 → four 1s → **P8 = 0**

Codeword: | Position | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | |---|---|---|---|---|---|---|---|---|---|---|---|---| | Bit | **1** | **0** | 1 | **0** | 0 | 1 | 0 | **0** | 1 | 1 | 1 | 1 |

Transmitted codeword = 1010 0100 1111 (12 bits).

> SUPPLEMENTARY convention warning: some textbooks (e.g. Stallings' *Data and Computer Communications*) number positions **from the right** and place data in a different order, which gives a different-looking codeword. Exam Q2(a) does not state a convention, so **write your convention in one line** ("positions numbered 1–12 from the left; check bits at 1, 2, 4, 8; data bits in order in 3, 5, 6, 7, 9–12") and the marker can follow your working. T2-2026 Q2 specifies exactly this convention.

★ Worked example HM2 (T2-2026 Q2(a)–(c), 8 marks): data 11001010

(a) Minimum check bits. Inequality **2^r ≥ m + r + 1** with m = 8. r = 3: 8 ≥ 12 ✘. r = 4: 16 ≥ 13 ✔. **r = 4.** Verify: $2^4 = 16 \geq 13$ ✔ while $2^3 = 8 < 12$.

(b) Placement and check bits: | Position | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | |---|---|---|---|---|---|---|---|---|---|---|---|---| | Type | P1 | P2 | D | P4 | D | D | D | P8 | D | D | D | D | | Data bit | | | **1** | | **1** | **0** | **0** | | **1** | **0** | **1** | **0** |

- **P1** covers 1, 3, 5, 7, 9, 11 → data 1, 1, 0, 1, 1 = four 1s → **P1 = 0**
- **P2** covers 2, 3, 6, 7, 10, 11 → data 1, 0, 0, 0, 1 = two 1s → **P2 = 0**
- **P4** covers 4, 5, 6, 7, 12 → data 1, 0, 0, 0 = one 1 → **P4 = 1**
- **P8** covers 8, 9, 10, 11, 12 → data 1, 0, 1, 0 = two 1s → **P8 = 0**

(c) Complete codeword (positions 1 → 12): | Position | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | |---|---|---|---|---|---|---|---|---|---|---|---|---| | Bit | **0** | **0** | 1 | **1** | 1 | 0 | 0 | **0** | 1 | 0 | 1 | 0 |

Transmitted codeword = 0011 1000 1010

How the receiver corrects an error (syndrome) SUPPLEMENTARY

Recompute each check over its group, including the received check bit. Write the failing checks as a binary number **C8 C4 C2 C1** (1 = fails). This number is the **position of the bad bit**; 0000 = no error.

Example using HM2: suppose bit 6 flips, so 0011 1**1**00 1010 is received.

- C1 (1, 3, 5, 7, 9, 11): 0, 1, 1, 0, 1, 1 = four 1s → OK → 0
- C2 (2, 3, 6, 7, 10, 11): 0, 1, **1**, 0, 0, 1 = three 1s → FAIL → 1
- C4 (4, 5, 6, 7, 12): 1, 1, **1**, 0, 0 = three 1s → FAIL → 1
- C8 (8, 9, 10, 11, 12): 0, 1, 0, 1, 0 = two 1s → OK → 0
- **Syndrome C8C4C2C1 = 0110 = 6** → flip bit 6 → corrected ✔

8.8 ★ Cyclic Redundancy Check (CRC)

Theory (the worksheet's worked example and links [ED]; supplementary explanation)

- Sender and receiver agree on a **generator G** of **r + 1 bits** (equivalently a **polynomial of degree r**), e.g. **$x^3 + 1$ ↔ 1001**, **$x^3 + x^2 + 1$ ↔ 1101**, **$x^3 + x + 1$ ↔ 1011**.
- **Polynomial ↔ bits**: write a 1 for every power present, from the highest power down to x^0. E.g. $x^3 + x + 1 = 1{\cdot}x^3 + 0{\cdot}x^2 + 1{\cdot}x^1 + 1{\cdot}x^0$ → **1011**.

Sender (the method used in the worksheet [ED]):

1. **Append r zeros** to the data (r = degree of G = number of bits in G − 1).
2. **Divide** by G using **modulo-2 long division** (XOR, no borrows).
3. The **r-bit remainder R is the CRC**.
4. **Transmit data + R** (R replaces the appended zeros). The transmitted frame is **exactly divisible by G**.

Receiver: divides the whole received frame by G. **Remainder 0 → accept; non-zero → error detected** ([ED]: *"Since the remainder is non-zero, we know that something went wrong!"*).

Modulo-2 division rules:

- If the leading bit of the current window is **1**, XOR it with G; if it is **0**, XOR with 0000 (i.e. just bring down the next bit).
- XOR: $0 \oplus 0 = 0$, $1 \oplus 1 = 0$, $0 \oplus 1 = 1 \oplus 0 = 1$.

★ Why modulo-2 arithmetic? (worksheet question [ED])

SUPPLEMENTARY: modulo-2 addition and subtraction are the **same operation (XOR)**, with **no carries or borrows**. This makes division:

- **simple and fast in hardware**: a CRC is just a **shift register with XOR gates**;
- **mathematically consistent**: polynomial arithmetic over GF(2), which gives CRC its strong detection properties (all single-bit errors, all double-bit errors with a suitable G, all odd-numbered errors if G has the factor (x + 1), and **all bursts of length ≤ r**).

Worksheet example CR0 [ED] (verified)

Message 11001001, $G = x^3 + 1 = 1001$.

- Append 000 → 11001001**000**; dividing by 1001 gives **remainder 011** (quotient 1101 0011).
- **Transmit 11001001 011.**
- If the **leftmost bit flips** → received 01001001011 → remainder **010** (the worksheet writes it as "10") → non-zero → **error detected** ✔.

★ Worked example CR1 (Exam Q2(b), 4 marks): data 101110, G = 1001

r = 3 → append **000** → dividend **101110000**

```
1 0 1 1 1 0 0 0 0
1 0 0 1                 ← leading 1: XOR with 1001
─────────
0 0 1 0 1 0 0 0 0
    1 0 1 0             ← next window starting with 1
    1 0 0 1
    ─────────
    0 0 1 1 0 0 0
        1 1 0 0
        1 0 0 1
        ─────────
        0 1 0 1 0
          1 0 1 0
          1 0 0 1
          ─────────
          0 0 1 1   ← remainder (last 3 bits) = 011
```

Step by step (each XOR shown on a 4-bit window):

1. 1011 ⊕ 1001 = 0010 → working value 001010000
2. 1010 ⊕ 1001 = 0011 → 000011000
3. 1100 ⊕ 1001 = 0101 → 000001010
4. 1010 ⊕ 1001 = 0011 → 000000011

Remainder R = 011. Transmitted bit string = 101110 011. Check: 101110011 ÷ 1001 leaves remainder 000 ✔.

★ Worked example CR2 (T2-2026 Q3, 6 marks): data 110101, G = 1011 ($x^3 + x + 1$)

(a) Degree 3 → append **3 zeros** → **110101000** (1 mark).

(b) Division:

1. 1101 ⊕ 1011 = 0110 → 011001000
2. 1100 ⊕ 1011 = 0111 → 001111000
3. 1111 ⊕ 1011 = 0100 → 000100000
4. 1000 ⊕ 1011 = 0011 → 000001100
5. 1100 ⊕ 1011 = 0111 → 000000111

```
110101000
1011
────
 1100            (0110 → bring down: 1100)
 1011
 ────
  1111
  1011
  ────
   1000
   1011
   ────
     1100        (0011 → bring down two bits: 1100)
     1011
     ────
      111        ← remainder
```

CRC remainder = 111.

(c) Transmitted bit string = **110101 111**. Check: 110101111 ÷ 1011 → remainder 000 ✔.

★ Worked example CR3 (T1B Q2, 16 marks): message 10011010, G = $x^3 + x^2 + 1$

(a) Polynomial → bits: $x^3 + x^2 + 0{\cdot}x + 1$ → **1101** (r = 3). Append 000 → **10011010000**.

Step	Window	⊕ 1101	Result	Working value
1	1001	1101	0100	01001010000
2	1001	1101	0100	00100010000
3	1000	1101	0101	00010110000
4	1011	1101	0110	00001100000
5	1100	1101	0001	00000001000
6	1000	1101	0101	00000000101

```
10011010000
1101
————
 1001
 1101
 ————
  1000
  1101
  ————
   1011
   1101
   ————
    1100
    1101
    ————
       1000      (0001 → bring down 3 bits → 1000)
       1101
       ————
        101      ← remainder
```

Remainder = 101 → **transmit 10011010 101**. Polynomial view: $M(x) = x^7 + x^4 + x^3 + x$; $x^3M(x) = x^{10} + x^7 + x^6 + x^4$, and the remainder mod $(x^3 + x^2 + 1)$ is $x^2 + 1 \leftrightarrow 101$ ✔.

(b) Leftmost bit inverted → received **00011010101**. Divide by 1101: the only step is 1101 ⊕ 1101 at positions 4–7 → 00000000101 → **remainder 101 ≠ 000** → the receiver knows an **error has occurred**, because a correct frame is **exactly divisible** by G (remainder 0). It then discards the frame and requests retransmission (ARQ).

Worked example CR4: the worksheet CRC-8 [ED] (verified)

- CRC-8 polynomial: **$x^8 + x^2 + x + 1$** → **100000111**.
- Data 1011 0011 0101 0110 + 8 zeros → remainder **1101 0101** → transmit **1011 0011 0101 0110 1101 0101** ✔ (worksheet result confirmed).
- **7th bit flipped** → received 1011 0**0**01 0101 0110 1101 0101 → remainder **0010 1010 (101010)** ≠ 0 → **error detected** ✔ (matches the worksheet image).

Worked examples CR5 (worksheet Kurose problems [ED], verified)

G = 10011 (r = 4). | D | Remainder R | Transmitted D·R | |---|---|---| | **1010101010** | **0100** | 1010101010 0100 | | (a) 1000100101 | **1000** | 1000100101 1000 | | (b) 0101101010 | **1111** | 0101101010 1111 | | (c) 0110100011 | **1110** | 0110100011 1110 |

Division for the main case D = 1010101010 (append 0000 → 10101010100000):

1. 10101 ⊕ 10011 = 00110 → 00110010100000
2. 11001 ⊕ 10011 = 01010 → 00010100100000
3. 10100 ⊕ 10011 = 00111 → 00000111100000
4. 11110 ⊕ 10011 = 01101 → 00000011010000
5. 11010 ⊕ 10011 = 01001 → 00000001001000
6. 10010 ⊕ 10011 = 00001 → 00000000000100 → **R = 0100** ✔

Worked example CR6 (worksheet [ED]): M(x) = 110001110110, G(x) = $x^5 + x^2 + x + 1$

- G = $x^5 + 0x^4 + 0x^3 + x^2 + x + 1$ → **100111** (r = 5) → append 00000.
- Steps (6-bit windows): 110001⊕100111 = 010110; 101101⊕100111 = 001010; 101010⊕100111 = 001101; 110111⊕100111 = 010000; 100000⊕100111 = 000111; 111000⊕100111 = 011111; 111110⊕100111 = 011001; 110010⊕100111 = 010101.
- **Remainder = 10101** → **T(x) = 110001110110 10101**.

8.9 Comparison of the four techniques (assembled)

Method	Redundancy	Detects	Corrects	Strength	Weakness
Single parity	1 bit	All **odd**-numbered errors (all single-bit)	No	Cheapest	Misses even-numbered errors (~50% of bursts)
2-D parity	Row + column bits	All 1-, 2-, 3-bit errors; many bursts	**Single-bit**	Simple correction	Misses some 4-bit rectangles
Internet checksum	One word (16 bits in IP/TCP; 4 in exercises)	Most errors	No	Simple software (addition)	Misses reordered words and compensating errors
Hamming	r bits, 2^r ≥ m + r + 1	Single (and double with an extra overall parity bit)	**Single-bit**	Forward correction, no retransmission	Weak against bursts
CRC	r bits (degree of G)	All single, all odd (with x + 1), **all bursts ≤ r**	No (detect + ARQ)	Very strong; simple hardware	Detection only

Where they appear in the module's technologies [BT]: Bluetooth uses **1/3-rate FEC** (send each bit 3 times, majority vote) on the 18-bit header, a **2/3-rate FEC** that is a **(15, 10) shortened Hamming code** (corrects all single and detects all double errors), an **8-bit HEC** on the header, a **16-bit CRC** on payloads, and **ARQ**. Full detail in Part 3.

8.10 Examination focus — Topic 8

Question	Paper	Marks
Hamming: 10101111, check bits + codeword	Exam Q2(a)	4
Hamming: 11001010 (inequality, placement, codeword)	T2-2026 Q2	8
CRC: 101110, G = 1001	Exam Q2(b)	4
CRC: 110101, G = 1011	T2-2026 Q3	6

Question	Paper	Marks
CRC: 10011010, $x^3 + x^2 + 1$ + leftmost bit flipped	T1B Q2	16
Checksum: 1001 1100 1010 0011	Exam Q2(c)	2
Checksum: 1010 1100 0110 1001 + receiver result	T2-2026 Q4	4
Single-bit vs burst; parity appropriateness	T1B Q1, T2-2026 Q5–6	10, 8

8.11 Topic 8 revision summary

- Detection: append check bits, recompute, compare. **ARQ** retransmits; **FEC** corrects.
- **Single-bit** = 1 bit changed; **burst** = ≥ 2 bits, length from first to last bad bit.
- **Parity** detects odd numbers of errors (all single-bit), misses even numbers (unreliable for bursts), **cannot correct**.
- **Checksum**: one's-complement sum with **end-around carry**, then **complement**. Receiver sum = **all 1s** if OK. Exam: **1011**; T2-2026: **1000** (receiver 1111).
- **Hamming**: **2^r ≥ m + r + 1** (8 data → 4 check bits, 12-bit codeword); P1, P2, P4, P8; syndrome = position of the error. 10101111 → **1010 0100 1111**; 11001010 → **0011 1000 1010**.
- **CRC**: append r zeros, XOR-divide by G, remainder = CRC; receiver remainder 0 = OK. 101110/1001 → **011**; 110101/1011 → **111**; 10011010/1101 → **101**.